

DIBYAJYOTI NAYAK

Ph.D. Candidate

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EDUCATION

Johns Hopkins University, Baltimore

Ph.D. in Scientific Machine Learning, **Advisor:** Prof. Somdatta Goswami, **CGPA:** 4.0/4.0

Jan 2025 – Present

Maryland, US

Indian Institute of Science, Bengaluru

M.Tech. (Research) in Computational and Data Sciences, **Advisor:** Prof. Konduri Aditya, **CGPA:** 9.5/10

Aug 2021 – Mar 2024

Karnataka, IN

National Institute of Technology, Tiruchirappalli

B.Tech. (Honors) in Mechanical Engineering, **Grade:** First Class with Distinction, **CGPA:** 9.34/10

Aug 2015 – May 2019

Tamil Nadu, IN

EXPERIENCE

GE Aerospace Research, Niskayuna

New York, US

Probabilistic Design Fellow Intern

May 2026 – Aug 2026

- Member of the Probabilistic Design and Materials Informatics (PDMI) team, developing uncertainty-aware, physics-informed surrogate models using probabilistic machine learning to accelerate numerical solvers.
- Designed deep learning-based warm starting strategies for high-fidelity CFD solvers, reducing solver iterations and accelerating convergence for complex flow simulations.
- Developed scalable geometric deep learning architectures as forward surrogate models for additive manufacturing, enabling rapid prediction of part quality and process outcomes.

HARMAN International, Bengaluru

Karnataka, IN

Research Intern

Jan 2024 – Sep 2024

- Worked in the Harman X research division on projects involving predictive modeling of mental state and vital signs estimation for driver monitoring systems.
- Explored distributed training strategies (DDP, FSDP), mixed precision, and model pruning to accelerate training of relevant CNN, GAN, and Transformer-based architectures on PyTorch.
- Optimized a Python-based code for large-scale data processing of camera video feed streams by leveraging multithreading, multiprocessing, vectorization, and efficient video read operations to achieve an overall speedup of 2.5x.
- Performed end-to-end data preprocessing (cleaning, wrangling, feature extraction) on raw camera video predictions; developed post-processing scripts for downstream statistical analysis.
- Developed a standardized interactive data visualization platform using Power BI and DAX, incorporating real-time data aggregation and filters to assess the impact of various metadata features on model predictions across different experiments.

Bajaj Auto Ltd., Pune

Maharashtra, IN

Assistant Manager (Engineering)

Jul 2019 – Jul 2021

- Developed internal gear teeth rolling technology for finishing of balancer gears for KTM and Pulsar models.
- Designed low-cost auto-deburring SPM unit for groove deburring in secondary gears for Pulsar and Platina models.
- Contributed to a project titled: “Standardization of special tools used in vehicle assembly” relevant to three-wheeler vehicles.
- Worked on eliminating the sizing process for powder metallurgy route gears for Pulsar and Avenger models.

Summer Intern (Engineering)

May 2018 – Jul 2018

- Developed a special gear-cutting cycle in the hobbing process to eliminate burr generation.

RELEVANT PROJECTS

Hybridizing Neural Operators and Solvers for Stable Long-Horizon Inference | Python, NumPy, scikit-learn, JAX, Flax, PyTorch

- Developed ANCHOR, an adaptive numerical-neural coupling framework that enhances existing sequential neural surrogates with targeted high-fidelity numerical corrections to enable stable, accurate, and reliable long-horizon predictions.
- Achieved robust, real-time error control by designing instance-aware, physics-informed residual monitoring, bounding error growth under 5-10% over long horizons and improving inference efficiency by up to 10x compared to standalone high-fidelity solvers on six canonical PDE benchmarks (1D/2D Burgers', 2D Allen-Cahn, 2D Cahn-Hilliard, 2D Navier Stokes, and 3D heat conduction).

Data-Efficient Time-Dependent PDE Surrogates: GNS vs NOs | Python, NumPy, scikit-learn, JAX, Flax, PyTorch

- Developed a Graph Neural Simulator (GNS) framework coupled with explicit numerical time-stepping schemes for PDE surrogate modeling, achieving under 1% test error using only 3% of training data, demonstrating superior generalization in limited-data regimes compared to neural operators (NOs).

- Conducted extensive evaluations across four canonical PDE systems, showing that the GNS surrogate achieves higher accuracy and stability over long inference horizons, reducing forecasting errors by over 82% and 99% compared to FNO- and DeepONet-based autoregressive baselines, respectively.

PITI-DeepONet: Temporal Tangent Space Operator Learning for High-Accuracy Inference | *Python, NumPy, scikit-learn, JAX, Flax*

- Proposed the Physics-Informed Time-Integrated Deep Operator Network (PITI-DeepONet), which employs a dual-output DeepONet architecture to learn the temporal derivative operator using either fully physics-informed or hybrid physics and data-driven objectives.
- Extensive evaluations across four canonical PDE systems demonstrate that PITI-DeepONet achieves enhanced accuracy and stability over long inference horizons, reducing forecasting errors by over 79% and 84% compared to full rollout and autoregressive methods, respectively.

TI-DeepONet: Learnable Time Integration for Stable Long-Term Extrapolation | *Python, NumPy, scikit-learn, JAX, Flax, PyTorch*

- Proposed the Time-Integrator embedded Deep Operator Network (TI-DeepONet), a framework that integrates neural operators with adaptive numerical time-stepping techniques to preserve the underlying Markovian structure of dynamical systems while substantially mitigating error propagation in extended temporal forecasting.
- Rigorous evaluation across six canonical PDE systems shows that the proposed framework achieves significant reductions in relative L_2 extrapolation error: approximately 96.3% compared to autoregressive implementations and 83.6% compared to fixed-horizon approaches.

A co-kurtosis PCA based dimensionality reduction with ANN reconstruction | *Python, NumPy, scikit-learn, TensorFlow, Keras*

- Proposed the CoK-PCA-ANN dimensionality reduction method, which combines a CoK-PCA-based projection (encoding) and an ANN-based reconstruction (decoding) to accurately capture stiff chemical dynamics in combustion datasets [1, 2].
- Investigated the efficacy of the CoK-PCA-ANN method in four combustion test cases spanning conventional single-stage to complex two-stage ignition kinetics, different combustion regimes, as well as from a simple homogeneous reactor to a spatiotemporally evolving 2D flow.

Extreme event detection in hydrogen combustion relevant to reheat burners | *Python, NumPy, scikit-learn*

- Employed a co-kurtosis tensor-based anomaly detection algorithm to identify extreme events such as flame instabilities in reheat burners of hydrogen-fired sequential gas turbine engines.
- Initial results were accepted as a short paper for oral presentation at ASPACC'23. A related work in collaboration with TU Delft has been published in the Proceedings of The Combustion Institute [3].

Use of Hidden Markov Models for keystroke biometric studies | *C++, Python, NumPy, Pandas, scikit-learn*

- Explored the use of a Partially Observable Hidden Markov Model (POHMM), an extension of the Hidden Markov Model (HMM), where both hidden states and emissions depend on an observable independent Markov chain, to classify users as imposters or genuine based on their keystroke dynamics.

Time-series traffic density detection and prediction | *Python, NumPy, Pandas, scikit-learn, TensorFlow, Keras*

- Modeled a sequential time series data of traffic density observed for 13 hours using a scalable encoder-decoder model based on Bi-LSTM network architecture.

Numerical Simulation of Solid-Solid Phase Change Material (SS-PCM) based heat sink | *MATLAB, SolidWorks, COMSOL Multiphysics*

- Performed a numerical heat transfer study on a heat sink embedded with layered perovskite (SS-PCM) to find an optimized fin configuration based on constant fin volume for different kinds of complex geometries such as circular, square, triangular, and tapered triangular cross-sections.

PUBLICATIONS

- Nayak, D. *et al.* A co-kurtosis PCA based dimensionality reduction with nonlinear reconstruction using neural networks. *Combustion and Flame* **259**, 113192. ISSN: 0010-2180 (2024).
- Aditya, K. *et al.* A co-kurtosis PCA based dimensionality reduction with neural network reconstruction for chemical kinetics in reacting flows. *Bulletin of the American Physical Society* (2023).
- Floris, M. *et al.* Data-driven identification of precursors of flashback in a lean hydrogen reheat combustor. *Proceedings of the Combustion Institute* **40**, 105524. ISSN: 1540-7489 (2024).
- Roy, R., Nayak, D. & Goswami, S. The Best of Both Worlds: Hybridizing Neural Operators and Solvers for Stable Long-Horizon Inference. *arXiv preprint arXiv:2512.19643* (2025).
- Nayak, D. & Goswami, S. Data-Efficient Time-Dependent PDE Surrogates: Graph Neural Simulators vs Neural Operators. *arXiv preprint arXiv:2509.06154* (2025).
- Mandl, L. *et al.* Physics-informed time-integrated DeepONet: Temporal tangent space operator learning for high-accuracy inference. *Computer Methods in Applied Mechanics and Engineering* **455**, 118917. ISSN: 0045-7825 (2026).
- Nayak, D. & Goswami, S. TI-DeepONet: Learnable time integration for stable long-term extrapolation. *Computer Methods in Applied Mechanics and Engineering* **456**, 118960. ISSN: 0045-7825 (2026).

ACADEMIC HONORS AND AWARDS

- Recipient of the NetApp Medal for Best M.Tech. (Research) Thesis, CDS, IISc, 2026.
- Recipient of the NVIDIA Academic Grant for research on AI-driven simulation of collective behavior in fish schooling.
- Selected for attending Google Research Week, 2023 with travel grant.
- Recipient of the Best Poster award at IISc CDS Alumni Meet, 2022.
- Recipient of the ANSYS M.Tech. (Research) Fellowship for the duration of 2021-2023.
- Runner-up, Indian Railway Design Challenge, Shaastra (annual technical festival, IIT Madras) in 2015.
- Recipient of the National Talent Search (NTS) Scholarship for the duration of 2011-2019.

RELEVANT COURSEWORK

- Probabilistic Methods in Mechanics
- Numerical Methods
- Deep Learning for Physical Systems
- Introduction to Data Science
- Applied Math for Engineers
- Machine Learning for Signal Processing
- Introduction to Computing for AI and ML
- Introduction to Scalable Systems
- Numerical Linear Algebra
- Stochastic Models and Applications
- Pattern Recognition and Neural Networks
- Operations Research

TECHNICAL SKILLS

Languages: Python, C++, C, MATLAB, SQL

Frameworks: NumPy, pandas, Matplotlib, scikit-learn, TensorFlow, Keras, PyTorch, JAX, Flax, MLflow, Optuna

Packages: Microsoft 365, Microsoft Power BI

CAD: SolidWorks, Siemens NX, PTC Creo

Simulation: COMSOL Multiphysics, Ansys, KISSsoft

TALKS / PRESENTATIONS

- **Towards Personalized Precision Medicine: AI-Accelerated Real-Time Laser Therapy Modeling**, Poster Presentation, DOM-WSE Research Retreat, JHU, Feb 2026
- **TI-DeepONet: Learnable Time Integration for Stable Long-Term Extrapolation**
 - Oral presentation, SIAM Conference on Uncertainty Quantification (UQ), Minneapolis, Mar 2026
 - Poster presentation, DSAI Symposium: AI in Daily Life, JHU, Sept 2025
 - Seminar, JHU-IITD SMaRT seminar series, JHU, Aug 2025
- **A co-kurtosis based dimensionality reduction with ANN based reconstruction of chemical kinetics in reacting flows**
 - Poster presentation, CDS Research Expo, IISc, Aug 2023
 - Oral Presentation, IGM, Aerospace Engineering, IISc, April 2023
 - Oral Presentation, EECS Students Research Symposium, IISc, April 2023
 - Oral Presentation, SIAM CSE, Amsterdam, Feb 2023
- **Characterization of ANN-based reconstruction for one-dimensional laminar premixed flames in the CoK-PCA framework**, Oral Presentation, Indo-German Conference on Computational Mathematics (IGCM), IISc, Mar 2023
- **Detection of extreme events in hydrogen combustion relevant to reheat burners**, Online Presentation, ASPACC, Taiwan, May 2023

TEACHING

Johns Hopkins University

Baltimore, US

Graduate Teaching Assistant

Jan 2026 – May 2026

- **EN.560.617.01.SP26:** Deep Learning for Physical Systems, offered by Prof. Somdatta Goswami, Department of Civil and Systems Engineering, JHU.
- Responsibilities include delivering theory lectures, conducting tutorials, evaluating project proposals, mentoring ongoing student projects, and designing and grading assignments and quizzes for a class of 30 students.

Indian Institute of Science

Bengaluru, IN

Graduate Teaching Assistant

Aug 2022 – Dec 2022

- **DS284:** Numerical Linear Algebra, offered by Prof. Phani Motamarri, Department of Computational and Data Sciences, IISc.
- Responsibilities included conducting tutorial sessions, framing and evaluating various assessments such as assignments, quizzes, and subjective exams for a class of 60 students.

EXTRACURRICULAR EXPERIENCE

- **Silver Medal** in Football competition at EECS Sports Meet, IISc, 2023
- **Gold Medal** in Carrom competition at Spectrum (Inter-departmental sports meet), IISc, 2023
- Football **Champions**, Horizons (Inter-departmental sports meet of NIT-T for sophomores), 2016
- Football **Champions**, Aaveg (Inter-hostel cultural festival of NIT-T for freshers), 2015

Society for Industrial and Applied Mathematics (SIAM) Student Chapter

Nov 2022 – Dec 2024

Member

IISc, Bengaluru

- Actively involved in organizing talks, workshops, industrial outreach and math outreach programmes to nearby schools/colleges.

Campus Placement Coaching (CPC), Department of Mechanical Engineering

Jan 2019 – May 2019

Subject Matter Expert

NIT, Tiruchirappalli

- Headed the technical team of the CPC committee which involved conducting written tests and technical interviews for the pre-final year undergraduate students to prepare them for upcoming placements.

Designers' Consortium (DC), Product Designing Club, NIT-T

Jul 2016 – May 2019

Project Manager

NIT, Tiruchirappalli

- **Facilitator**, Workshop on Modeling in Creo in Synergy (National Technical Symposium, Mechanical Engineering, NIT-T).
- **Facilitator**, Workshop *Kinedes* on Design and Fabrication of a Pantograph and Rocker-Bogie Mechanism for freshers.
- **Event Manager**, Pragyan Design Challenge, Pragyan (International Techno-Management Festival, NIT-T)

Workshops and Informals Team, Festember, NIT-T

Apr 2016 – Sep 2016

Coordinator

NIT, Tiruchirappalli

- Conducted various cultural workshops and informal events as a part of Festember (National Cultural Festival of NIT-T).

REFERENCES

- Prof. Somdatta Goswami, Assistant Professor, Department of Civil and Systems Engineering, Johns Hopkins University, Baltimore, MD, US
- Prof. Kami Mohammadi, Assistant Professor, Department of Civil and Environmental Engineering, The University of Utah, Salt Lake City, UT, US